

G S E F O R E C L O S U R E

Disposition Analysis

Predicting Third-Party Sale vs REO Reversion at Auction

April 2026 | Technical Briefing

Executive Summary

0.80

Test ROC-AUC (random split)

10.2×

Top-decile / bottom-decile sale-rate spread

\$63–128M

Projected annual value

1

Loan fundamentals and market context dominate

Top 5 drivers (state, equity, BPO discount, LTV, property type) account for ~60% of feature importance. The model identifies high-confidence sales before any auction marketing begins.

2

Auction venue is not a meaningful lever

Judicial vs non-judicial: 0.4pp gap. Online vs offline (indirect test): 3.4pp raw, but collapses to 0.18pp after controlling for market context — a market-selection effect, not a venue effect.

3

Decile-based triage is the operational lever

Bottom-2 deciles carry 84% REO rate. Targeted intervention there (early loss mitigation, mod-to-market) at 5–15% effectiveness yields \$33–98M, plus \$30M from top-decile efficiency.

PROBLEM STATEMENT

The problem we set out to solve

When a foreclosed property goes to auction, two outcomes are possible:

OUTCOME A

3rd-party sale at auction

External bidder (typically an investor) buys the property at auction. Bank recovers value quickly, minimal carrying cost. Clean exit.

OUTCOME B

Reverts to REO

No qualifying bid received. Property reverts to lender ownership — bank now holds and manages real estate it never wanted. Estimated all-in cost: \$50K–80K per property.

THE QUESTION

Given a foreclosure case at intake, can we predict which outcome will occur — and use that prediction to triage operations and intervene before auction?

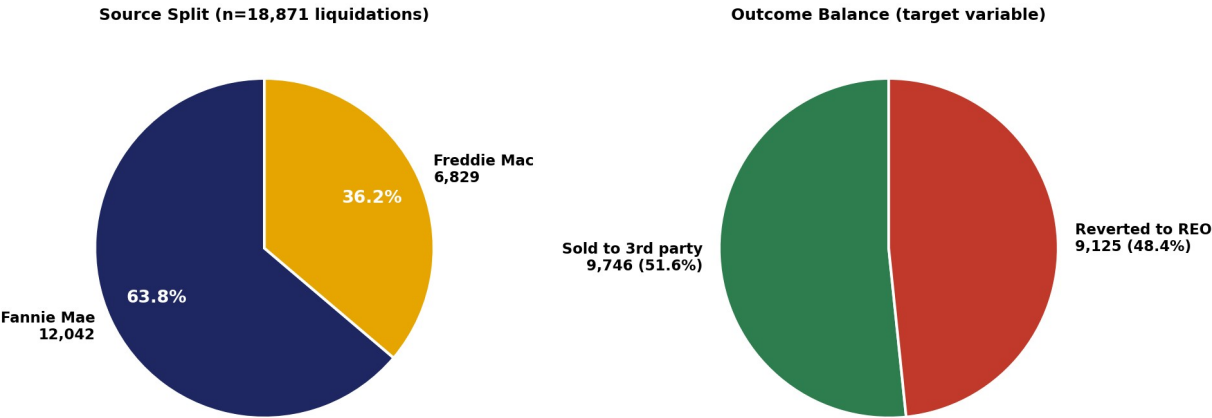
Data foundation

18,871 GSE foreclosure liquidations
2016–2024 dispositions

49 Quarterly source files
Fannie + Freddie loan-performance

30+ External data sources
FRED, FHFA, Census, Zillow, BLS, IRS

57 Engineered features
52 numeric + 5 categorical



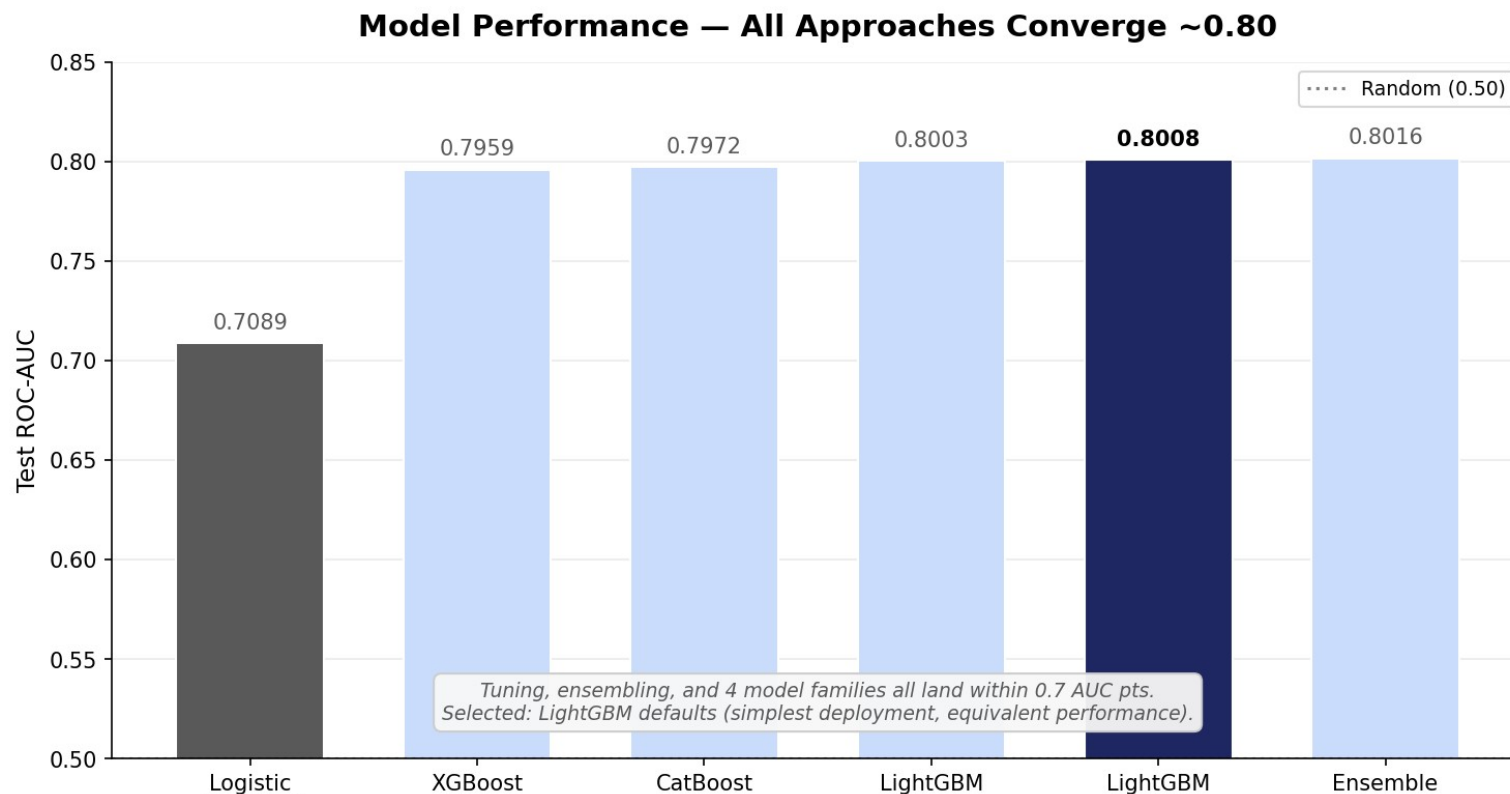
Source split (left) and target balance (right)

Why 30+ external sources?

Loan-only models reach AUC ~0.71. The remaining performance comes from external context — what investors see when deciding to bid. Sources are organized into 8 categories aligned with the investor decision framework.



Model performance — 0.80 AUC, validated across approaches



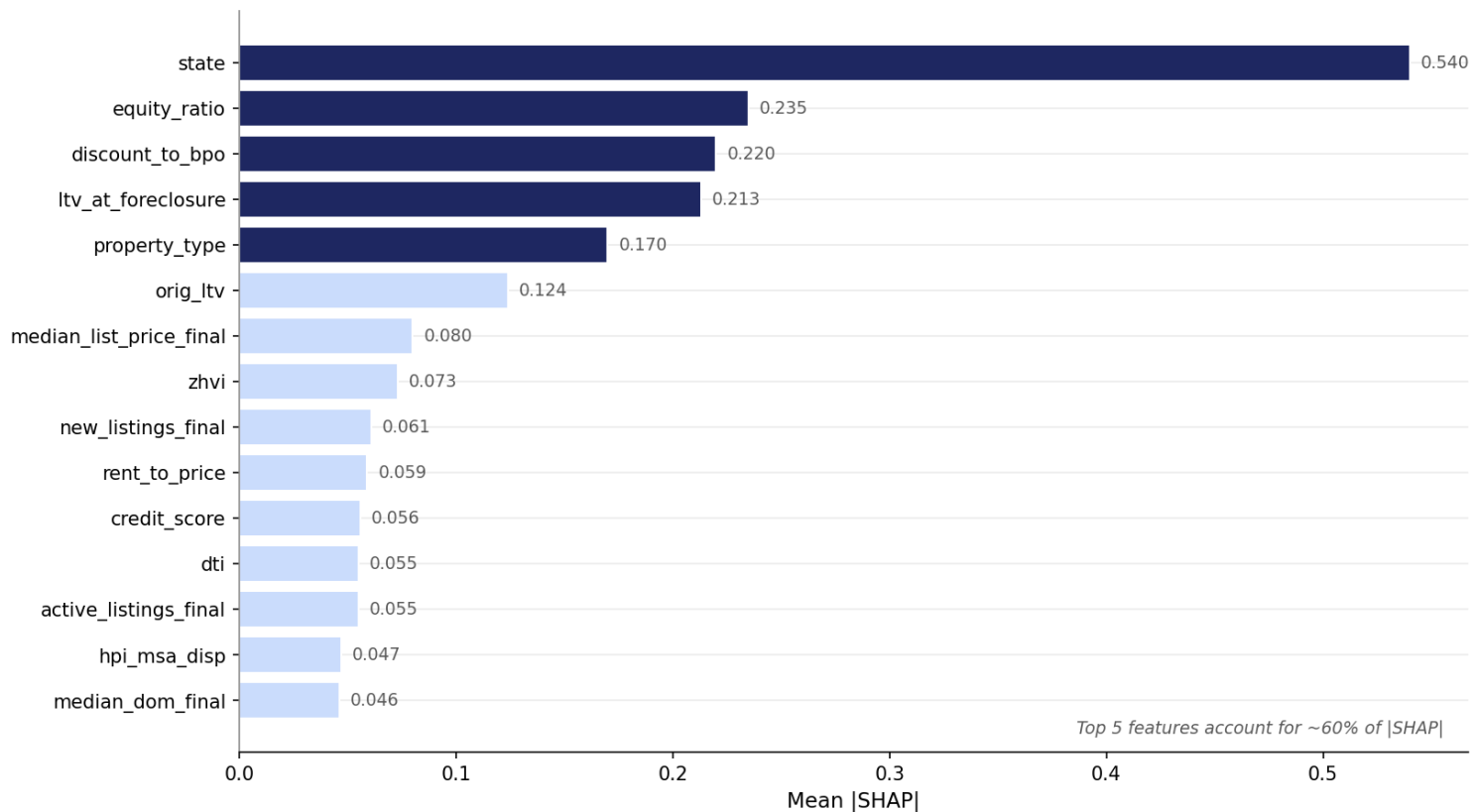
What this means

- All four GBT libraries land within 0.7 AUC pts of each other.
- 40-trial Optuna tuning produced no improvement — defaults were already optimal.
- Ensembling: +0.0008 AUC. Not worth 3× deployment overhead.
- Selected: LightGBM with default hyperparameters. Simplest, equivalent.

Time-aware split (train 2016-22, test 2023-24): AUC drops to 0.7365 — sensitivity to macro regime shift. *Argues for quarterly retraining.*

What drives whether a property sells?

Top 15 Drivers of Sale Probability (SHAP)



Top 5 drivers explain ~60%

1

State

AZ/UT/NV: 75-81% sell. Late-cycle Midwest: <40%.

2

Equity ratio

More equity → investor will bid.

3

Discount to BPO

Auction starting price relative to market value.

4

LTV at foreclosure

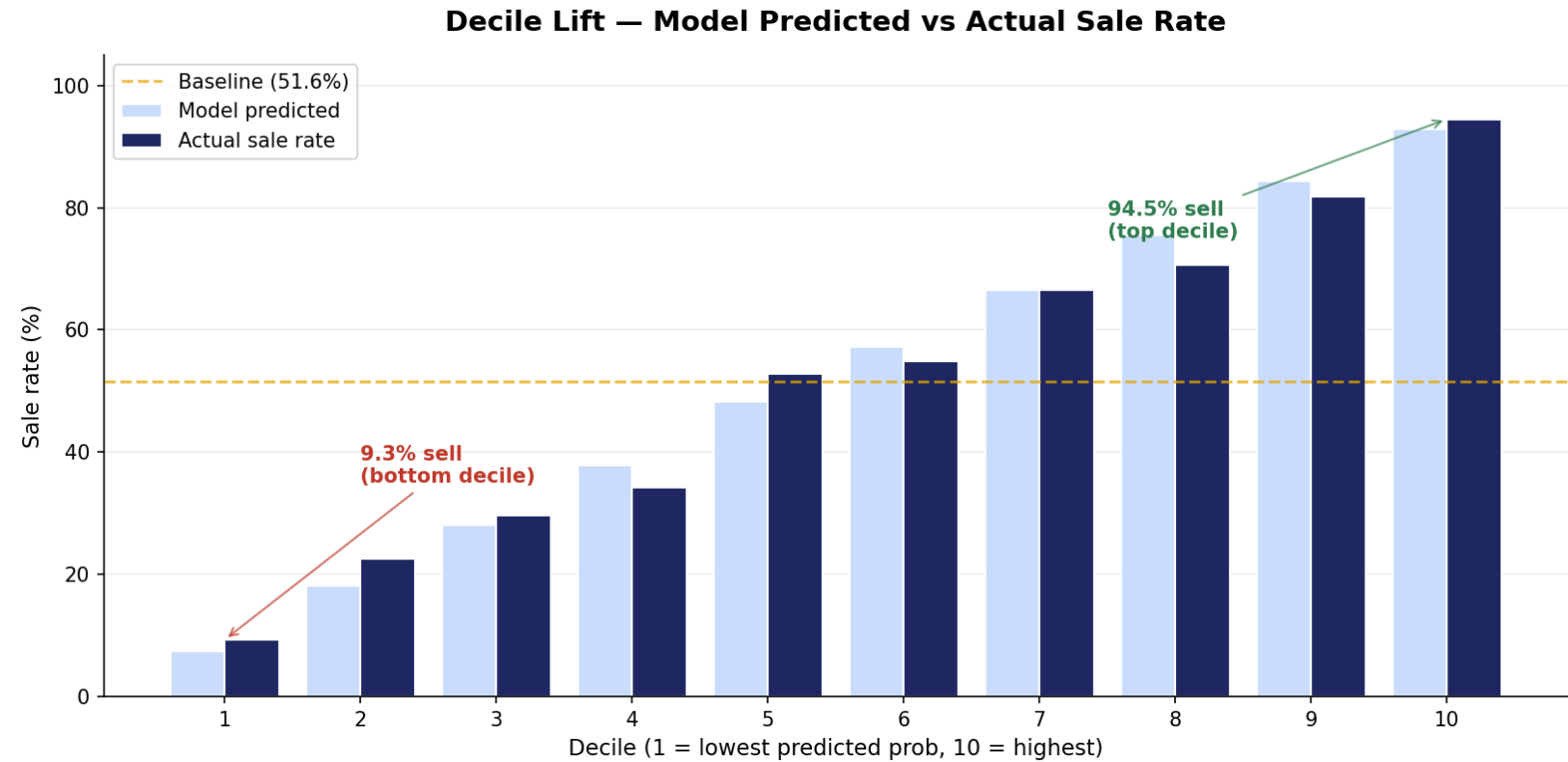
Debt overhang — high LTV → REO.

5

Property type

SFR sells more than condo or manufactured.

The operational lever — decile-based triage



How to use this

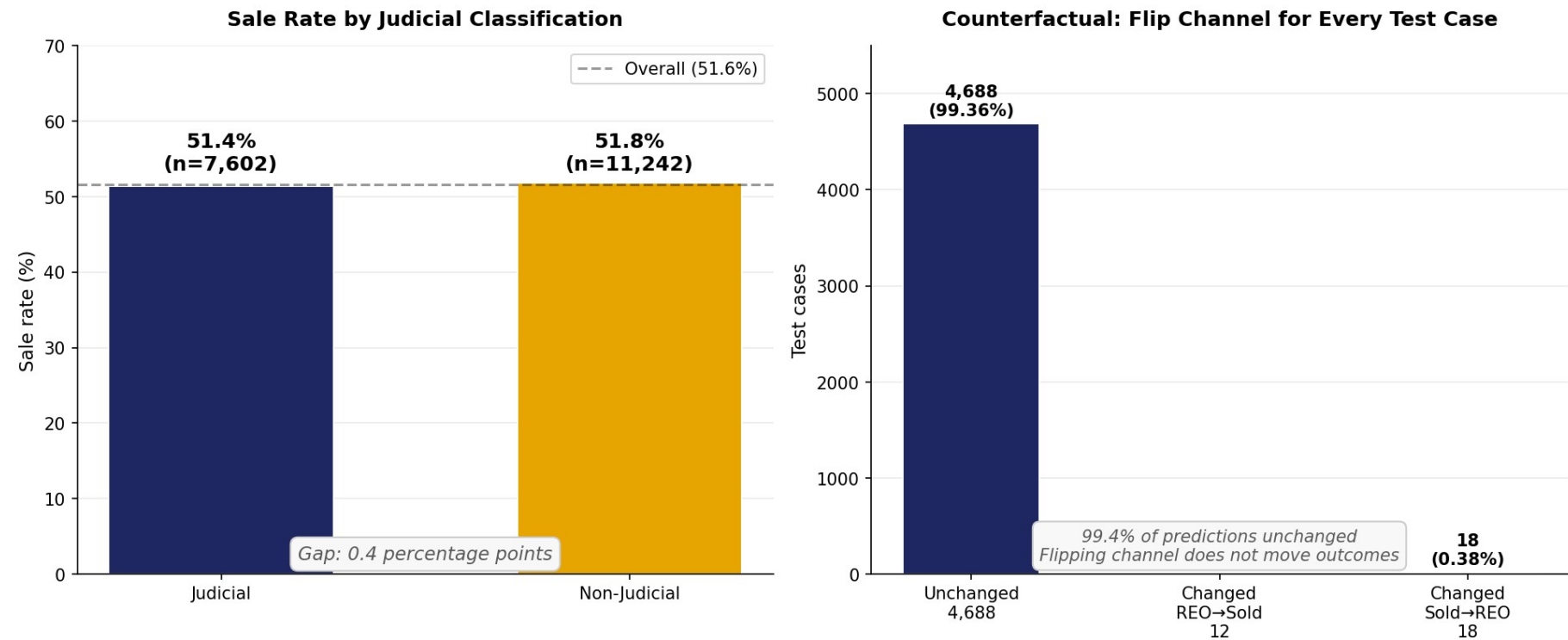
- TOP 2 DECILES**
88–95% sell rate. Auto-process. Minimal marketing.
- MIDDLE 6 DECILES**
Standard protocol. Most uncertainty, where effort moves the needle.
- BOTTOM 2 DECILES**
9–23% sell rate. Active intervention. Begin REO ops prep.

10.2× spread between top decile (94.5%) and bottom decile (9.3%) sale rates.

Does the auction venue matter? — Test 1

Test 1: Judicial vs Non-Judicial state classification

Does the state's legal foreclosure framework drive disposition outcome?

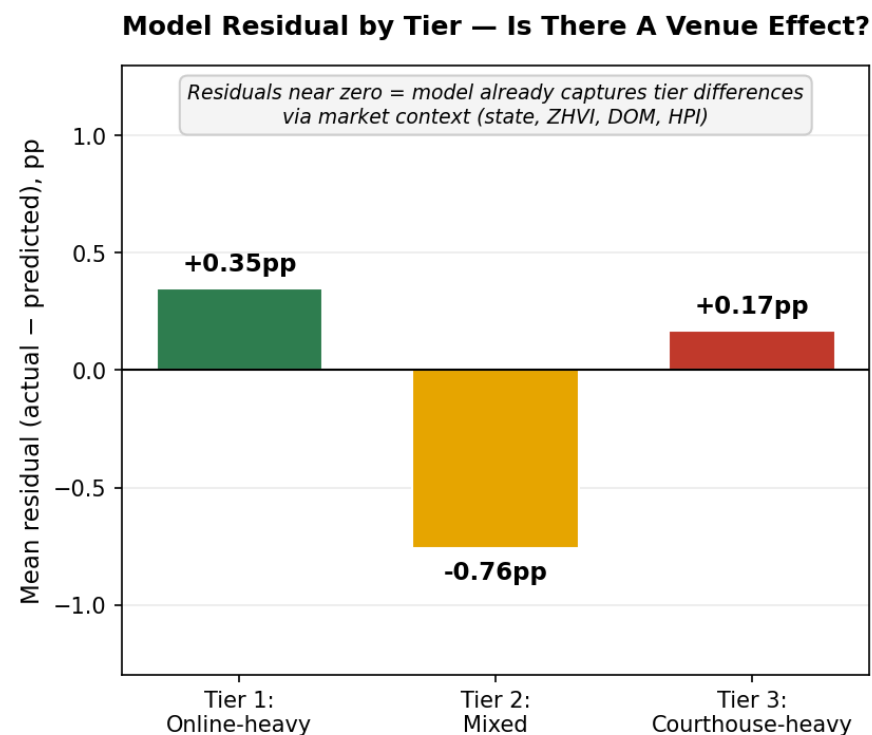
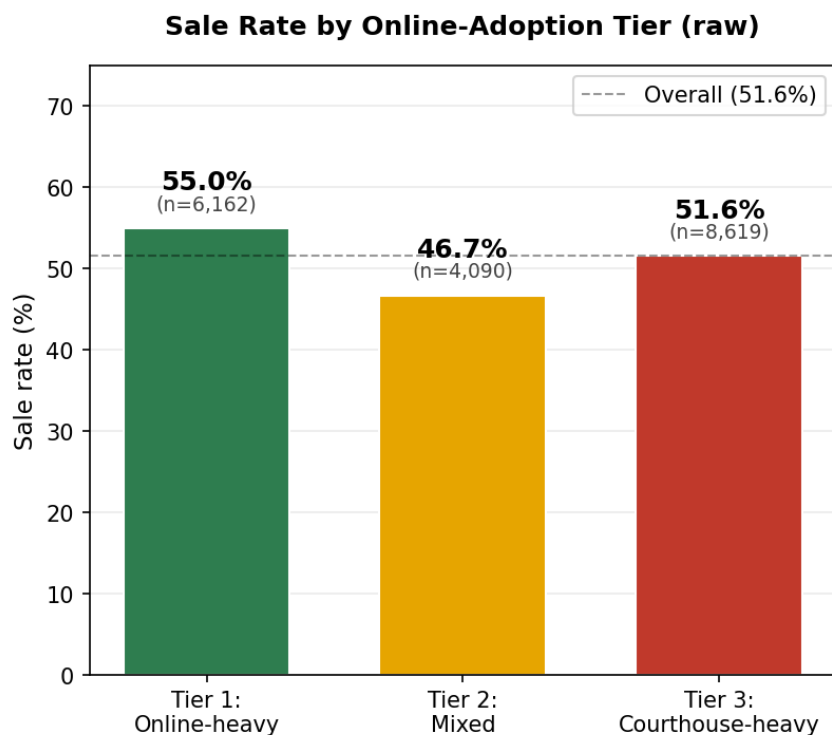


Finding: 0.4pp gap between judicial and non-judicial. Counterfactual flip changes only 12 of 4,718 predictions (0.25%).

Does the auction venue matter? — Test 2

Test 2: Online vs offline (indirect, via state platform-adoption tier)

GSE data does not record actual venue. We grouped states by online-platform adoption and compared raw rates AND model residuals.



Finding: Raw 3.4pp gap collapses to 0.18pp once market context is controlled for. The apparent venue effect is market-selection.

KEY FINDINGS

Venue is not the lever

0.4 pp

Judicial vs Non-Judicial sale rate gap

A statistical wash; selection effect across states.

0.25%

Of cases would change outcome if channel was flipped

12 of 4,718 test predictions move; 99.4% unchanged.

3.4 → 0.18 pp

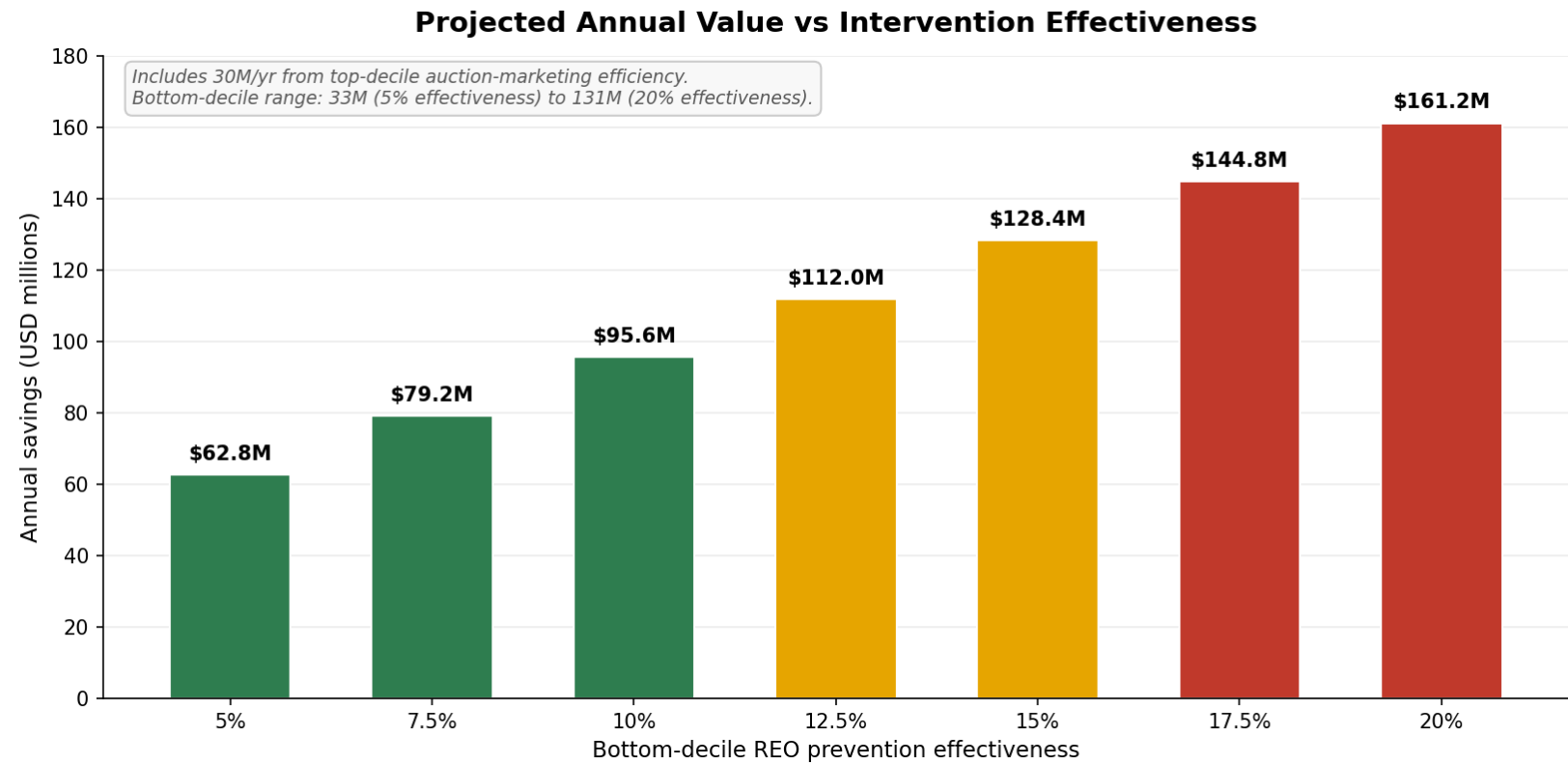
Online-vs-offline raw → residual gap

After market context, the gap is 95% explained away.

WHAT WE CANNOT DEFINITELY ANSWER

GSE data does not record actual auction venue. Our online-vs-offline test uses state-level platform-adoption tiers as a proxy. To test venue rigorously requires -internal auction-venue records joined to disposition outcomes — a 4-6 week analytical project.

Sizing the opportunity



Combined annual upside

Low (5% effectiveness)

\$62.8M

Mid (10%)

\$95.6M

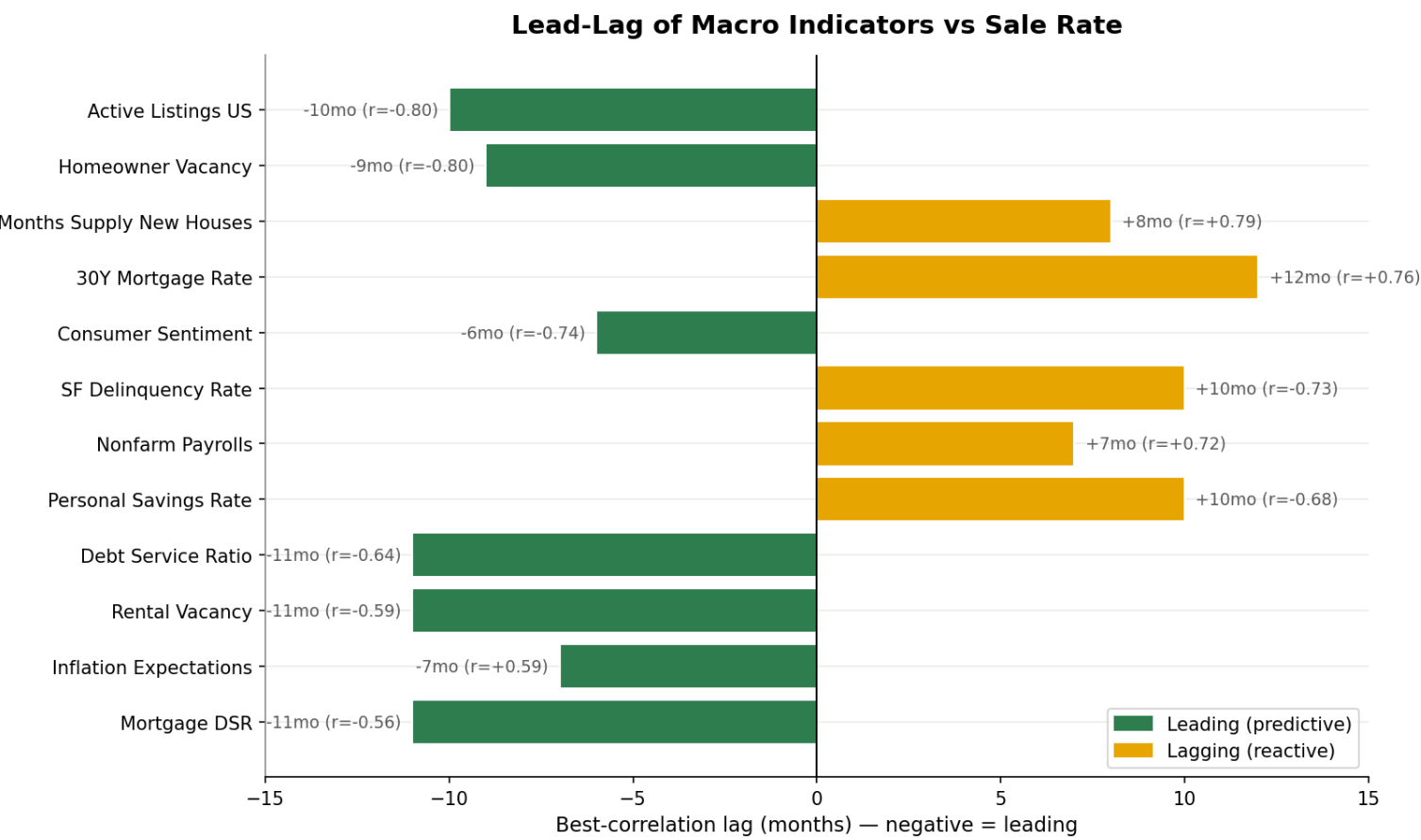
High (15%)

\$128.4M

Includes \$30M/yr from top-decile auction-marketing efficiency. Bottom-decile component scales linearly with intervention success rate.

Assumes: \$65K/REO holding cost · 200K US GSE foreclosures/yr · 30% share · 84% bottom-decile REO rate. Replace with internal numbers before commitment.

Macro indicators that lead the sale rate



Forward indicators

Six macro signals move 6–11 months BEFORE the disposition outcome:

- Active US listings
- Homeowner vacancy
- Consumer sentiment
- Debt service ratio
- Rental vacancy
- Inflation expectations

Operationally

Use as forward signals to scale bottom-decile triage staffing 6–11 months ahead of REO-rate spikes.

Two real test-set examples

CASE A — HIGH-CONFIDENCE SALE

Predicted: 0.986

Actual: SOLD ✓

Profile

- Florida, owner-occupied
- \$84K loan, originated 2016, 80% LTV
- Disposed April 2022

Why the model said SELL

- LTV at foreclosure: 0.43 (heavy de-leverage via appreciation)
- BPO discount: 54.9% (deep auction starting price)
- Hot Florida ZIP, ZHVI \$546K, 26-day metro DOM

CASE B — HIGH-CONFIDENCE REO

Predicted: 0.011

Actual: REVERTED TO REO ✓

Profile

- Minnesota, owner-occupied
- \$167K loan, originated 2018, 84% LTV
- Credit 799, DTI 48 (high stress at origination)

Why the model said REVERT

- Minnesota historically low sale rate (29%)
- Persistent debt stress (high origination DTI)
- Disposition-period market unfavorable

From research to production — how this becomes a 2026 capability

The model can score any 2026 case at intake today. Two layers of work turn it into a production capability.

LAYER 1

Scoring pipeline

6–11 weeks of engineering

- Intake adapter** — pulls new cases from foreclosure system
- Feature joiner** — pulls fresh macro + market context
- Scoring service** — REST API or batch job for ops systems
- Monitoring** — score drift detection, feature coverage alerts
- UI integration** — surfaces score and decile to case manager

Result: every 2026 case gets a probability of sale and decile assignment at intake.

LAYER 2

Quarterly retraining

3–5 days per quarter

- Pull new GSE disposition data** (Fannie + Freddie quarterly)
- Refresh external sources** (FRED, Zillow, ACS, Realtor.com)
- Rebuild feature matrix** (automated pipeline)
- Retrain LightGBM** with same hyperparameters
- Validate on holdout** before promoting to production

Without retraining: AUC drifts from 0.80 → 0.74 in one year. With it: stays at 0.80.

Recommendations

SHORT-TERM

next 30–90 days

- Validate cost assumptions internally — replace \$65K/REO and 30% market share with actuals
- Pilot decile triage on 6–8 weeks of incoming cases without operational change
- Identify the BPO data integration owner — highest-leverage data improvement

MEDIUM-TERM

3–6 months

- Productionize the model with quarterly retraining
- Stand up a triage workflow tied to the score
- Replace BPO proxy with real BPO data (+0.02–0.04 AUC improvement)

LONG-TERM

6–12 months

- Test online-vs-offline rigorously with internal venue records
- Extend to non-GSE portfolios (FHA, VA, private-label)
- Add full property-characteristic data (ATTOM, CoreLogic, MLS)

THREE THINGS TO REMEMBER

Foreclosure outcomes are predictable —
and the lever is loan + market, not auction venue.

01

AUC 0.80

on a model that achieves 10× lift between top and bottom deciles.

02

Venue isn't the lever

0.4pp judicial gap; 0.18pp residual online-vs-offline.

03

\$63–128M annual upside

from decile-based triage on existing caseload.

Discussion • Q&A